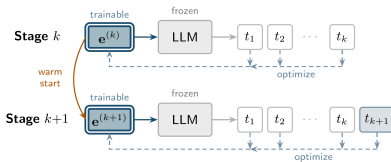


1 Method

Optimise a **per-sample** input embedding $\mathbf{e}$ so that a **frozen LM** reconstructs a text. The target grows **one token at a time**; the next stage starts only after **perfect** greedy reconstruction of the current prefix.



- Halting on the first failure gives a **sharp capacity boundary** per sample.
- **100% greedy reconstruction** at every accepted stage.
- Saving $\mathbf{e}$ after each stage yields the **optimisation trajectory**.

2 Capacity

Longest text a frozen model reconstructs perfectly from a **single** embedding, with and without a low-dimensional projection:

Model	Tokens	+LowDim	Boost
Llama-3.1-8B	1438	1697	1.2×
Pythia-1.4B	430	500	1.2×
SmolLM2-1.7B	335	957	2.9×
Gemma-3-4B	214	697	3.3×

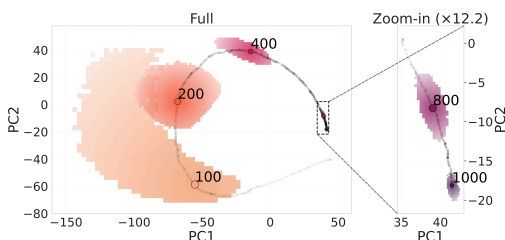
Low-dim projection

Optimise $\mathbf{z}$ with a per-sample affine map $\mathbf{W}, \mathbf{b}$:

$$\mathbf{e} = \mathbf{W}\mathbf{z} + \mathbf{b} \in \mathbb{R}^d, \quad \mathbf{z} \in \mathbb{R}^k, \quad k \ll d$$

3 Trajectories

- Trajectories are **intrinsically low-dimensional**: 30–100 PCA components explain 99% of the variance, although they live in $\mathbb{R}^{4096}$.
- Accuracy basins **shrink** with sequence length — longer texts leave less room for error.



Llama-3.1-8B trajectory on PC1–PC2 (65.7% of variance); coloured regions are local accuracy basins.

4 Understanding

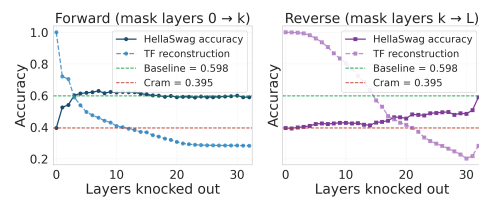


Prepending a crammed embedding **drops** benchmark accuracy even with the original prefix in context:

Model	HS base	HS +emb	ARC base	ARC +emb
Pythia-1.4B	44.0%	37.6%	53.1%	49.1%
SmolLM2-1.7B	53.6%	37.0%	67.3%	56.0%
Llama-3.1-8B	61.9%	40.8%	63.5%	43.3%

On **generative 5-shot MMLU** accuracy falls to **≈0%** with the optimised embedding, while a random-embedding control stays at the baseline.

5 Attention knockout

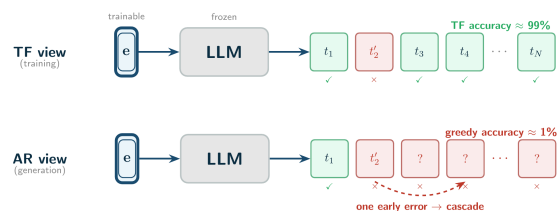


Llama-3.1-8B. Forward knockout (masking early layers first) restores downstream accuracy, while reverse knockout fails until the early layers are masked too.

Early layers drive both the reconstruction and the capability collapse; the 20–60% late-layer attention mass is a symptom. The pattern holds for Llama-3.1-8B, Pythia-1.4B and SmolLM2-1.7B.

6 Full cramming

Full cramming fixes the **number of tokens** and optimises them jointly; progressive cramming fixes **accuracy at 100%** and grows the text. Full cramming reaches more tokens, but its residual errors sit at **early positions**, and one early miss cascades through greedy generation:



Model / type	Tokens	TF acc.	Greedy
Llama-8B / full	1568	99.96%	40.4%
Llama-8B / progr.	1353	100%	89.8%
Pythia / full	512	99.71%	44.2%
Pythia / progr.	390	100%	96.9%

